Designing Robots 🛠️

Brownie Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

Robots have JOBS! 💼

Robots help people every day: - 🏠 Vacuum robots clean floors - 🚀 Rovers explore Mars - 🏭 Robot arms build cars - 🏥 Robots deliver medicine in hospitals - 🐕 Robot dogs check dangerous places

What job do YOU wish a robot could do for you?

Card Sort Game 🃏

Sort the robot cards into teams: - Robots that help at **home** - Robots that **explore** - Robots that work in **factories** - Robots that keep people **safe**

Sense → Think → Act 🧠

Every robot works the same way:

SENSE 👀	THINK 🧠	ACT 🦾
“I see a wall!”	“I should turn!”	*turns around*
“It’s dark!”	“Lights needed!”	*lights on*

Let’s play it! Your mentor calls out what the robot senses — YOU act out what it does!

Engineers design FIRST

Before building, engineers answer three questions:

1. What does my robot **SENSE**?
2. What does my robot **THINK** about?
3. What does my robot **DO**?

Your turn: Build a Robot Prototype! 

1. Pick a problem: messy room? thirsty plants? heavy backpack?
2. Fill out your design sheet:
   - My robot's **name**
   - My robot's **job**
   - What it **senses**
   - What it **does**
3. Build it with craft supplies!

A **prototype** is a first model — it doesn't have to really work!

Gallery Walk 🖼️

- Put your robot on the table
- Walk around and look at everyone's robots
- Leave ONE sticky-note compliment on someone else's robot 💌

You did it! 🎉

Today you: - ✅ Learned the jobs robots do - ✅ Played Sense → Think → Act - ✅ Designed a robot that solves a problem - ✅ Built a prototype like a real engineer

You are an engineer now!

Keep going at home! 🏠

- Watch Mars rover videos at **nasa.gov**
- Design more robots with paper and craft supplies
- **Tinkercad** (tinkercad.com) — design in 3D for free
- Next badge: **Showcasing Robots** — share your design with the world!